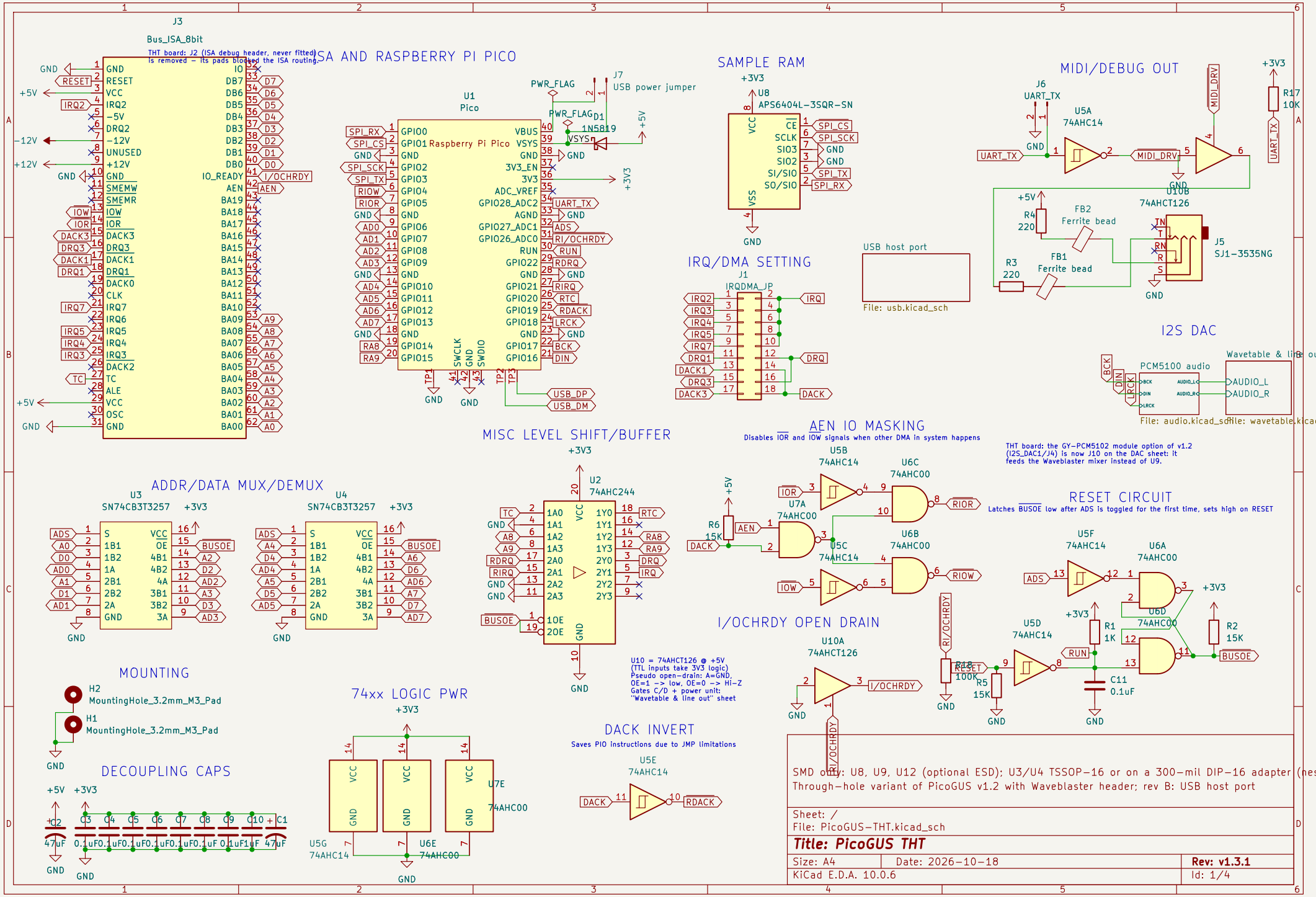
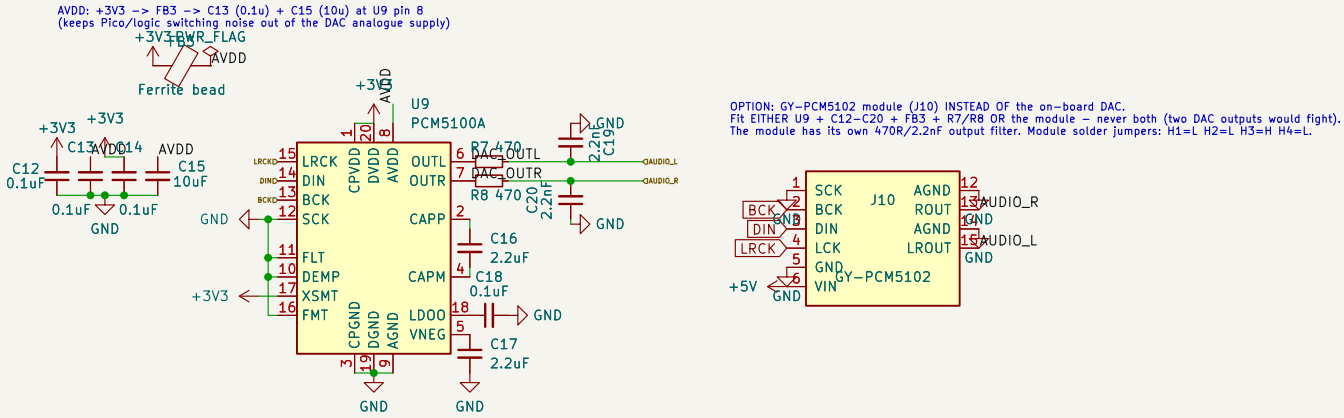

ISA AND RASPBERRY PI PICO
J3
Bus_ISA_8bit
THT board: J2 (ISA debug header, never fitted) is removed — its pads blocked the ISA routing.
U1
Pico
Raspberry Pi Pico
J7
USB power jumper
PWR_FLAG
D1
1N5819
SAMPLE RAM
U8
APS6404L-3SQR-SN
MIDI/DEBUG OUT
J6
UART_TX
U5A
74AHC14
U10B
74AHCT126
MIDI_DRV
R17
10K
R4
220
R3
220
FB2
Ferrite bead
FB1
Ferrite bead
J5
SJ1-3535NG
USB host port
File: usb.kicad_sch
IRQ/DMA SETTING
J1
IRQDMA_JP
I2S DAC
PCM5100 audio
Wavetable & line out
AUDIO_L
AUDIO_R
File: audio.kicad_sch
File: wavetable.kicad_sch
MISC LEVEL SHIFT/BUFFER
U2
74AHC244
AEN IO MASKING
Disables IOR and IOW signals when other DMA in system happens
U5B
74AHC14
U6C
74AHC00
U7A
74AHC00
U6B
74AHC00
U5C
74AHC14
THT board: the GY-PCM5102 module option of v1.2 (I2S_DAC1/J4) is now J10 on the DAC sheet: it feeds the Waveblaster mixer instead of U9.
RESET CIRCUIT
Latches BUSOE low after ADS is toggled for the first time, sets high on RESET
U5F
74AHC14
U6A
74AHC00
U6D
74AHC00
U5D
74AHC14
R1
1K
R2
15K
R5
15K
R18
100K
C11
0.1uF
ADDR/DATA MUX/DEMUX
U3
SN74CB3T3257
U4
SN74CB3T3257
I/OCHRDY OPEN DRAIN
U10A
74AHCT126
U10 = 74AHCT126 @ +5V (TTL inputs take 3V3 logic) Pseudo open-drain: A=GND, OE=1 -> low, OE=0 -> Hi-Z Gates C/D + power unit: "Wavetable & line out" sheet
MOUNTING
H2
MountingHole_3.2mm_M3_Pad
H1
MountingHole_3.2mm_M3_Pad
DECOUPLING CAPS
74xx LOGIC PWR
U7E
74AHC00
U5G
74AHC14
U6E
74AHC00
DACK INVERT
Saves PIO instructions due to JMP limitations
U5E
74AHC14
SMD only: U8, U9, U12 (optional ESD); U3/U4 TSSOP-16 or on a 300-mil DIP-16 adapter (nested fo
Through-hole variant of PicoGUS v1.2 with Waveblaster header; rev B: USB host port
Sheet: /
File: PicoGUS-THT.kicad_sch
Title: PicoGUS THT
Size: A4
Date: 2026-10-18
Rev: v1.3.1
KiCad E.D.A. 10.0.6
Id: 1/4



U9 stays SMD (TSSOP-20): PCM5100A/5101A/5102A only exist as SMD

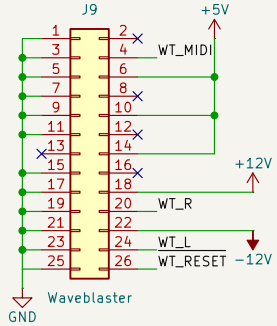
Sheet: /PCM5100 audio/
File: audio.kicad_sch

Title: PicoGUS THT - PCM510x DAC

Size: A4	Date: 2026-10-18	**Rev: v1.3.1**
KiCad E.D.A. 10.0.6		Id: 2/4

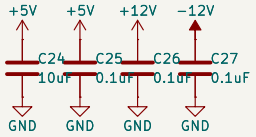
WAVEBLASTER HEADER

2x13 male header, 2.54 mm. Daughterboard (female socket) sits over the card; footprint has the standard standoff holes.



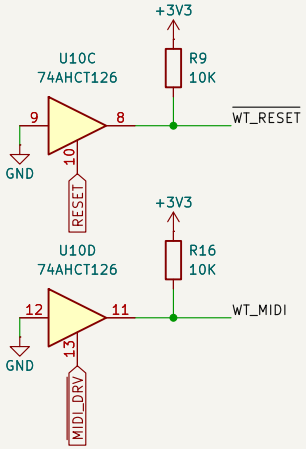
DGND 1–11, AGND 15–25: both to the ground plane

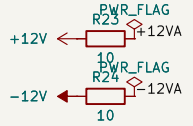
WT SUPPLY DECOUPLING (place at J9)



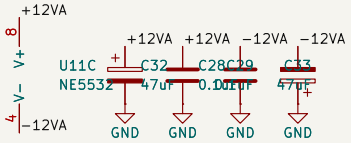
WT CONTROL (U10 = 74AHCT126 @ +5V)

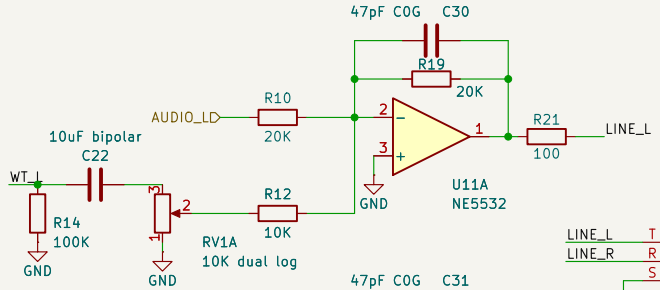
Both outputs are pseudo open-drain (A=GND, OE drives) with pull-ups to +3V3, so the daughterboard never sees 5 V on /RESET or MIDI (some DBs have 3.3 V inputs). /RESET: low while ISA RESET is high. MIDI: same data as the MIDI-out jack (MPU-401 UART).

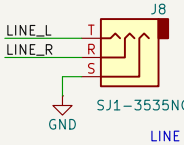




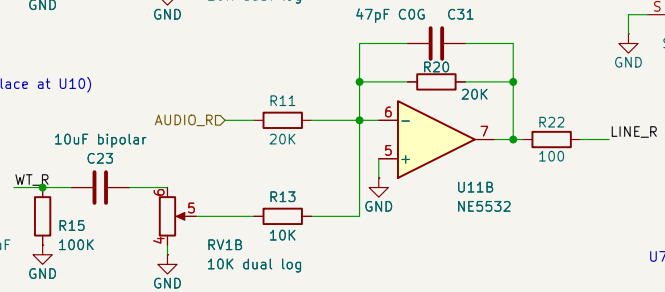
U11 SUPPLY: ISA +/−12 V via R23/R24 (10R) + C32/C33 (47u), C2



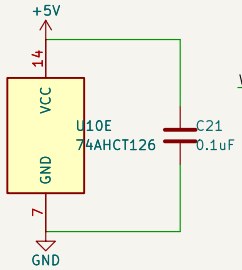




LINE OUT (green)



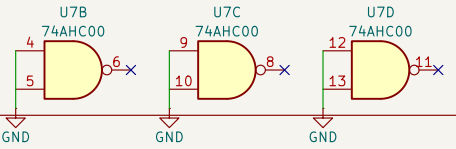
U10 POWER (+5V) + DECOUPLING (place at U10)



WAVETABLE VOLUME + MIXER

RV1 (dual 10k log) sets the WT level: pins 3/6 = CW end (signal), 1/4 = CCW end (GND). U11 is an inverting summer: PCM 20k in / 20k fb (−unity, −0.2 dB), WT 10k in (x2 = +6 dB at RV1 max; DB outputs are ~1–1.2 Vrms vs 2.1 Vrms PCM). The WT knob does not change the PCM level. C22/C23 (bipolar) block DC from the DB, R14/R15 keep them discharged when no DB is fitted. Output: 100R, DC-coupled like v1.2.

U7 SPARE GATES (74AHC00)

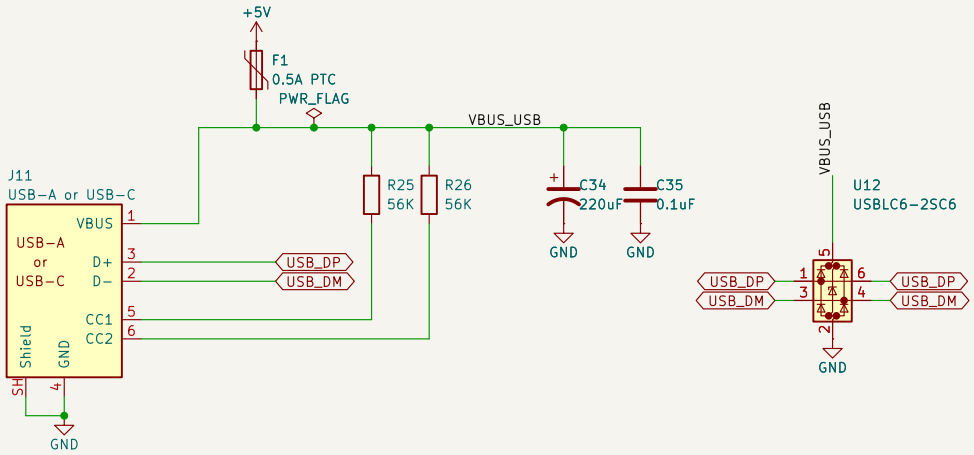


Derived from the hand386 ISA adapter mixer and the PicoGUS 2.0 WT header
Waveblaster (wavetable) daughterboard header, analog WT volume and op-amp mixer

Sheet: /Wavetable & line out/
File: wavetable.kicad_sch

Title: PicoGUS THT — Waveblaster header & line-out mixer

Size: A4	Date: 2026-10-18	**Rev: v1.3.1**
KiCad E.D.A. 10.0.6		Id: 3/4



USB HOST PORT (rev B)

Data straight from the Pico's USB test points TP3 (D+) / TP2 (D−) (U1, sheet 1), behind the 27R series resistors on the Pico module. The Pico must be soldered FLAT on the board (castellated edge or through-hole pins, no header) so TP1–TP3 can be soldered from the back of the card.
With the Pico on headers or in a socket the TP pads do not reach the card: wire TP1–TP3 to the holes, or use a micro-USB OTG adapter on the Pico (with jumper J7) as on v1.3.0. Never use both at once.
VBUS from ISA +5V via F1 (resettable, 0.5 A hold). The Pico VBUS pin is not used for this port.
J11 takes a USB-A OR a USB-C receptacle. R25/R26 (Rp 56k = default USB power) tell a USB-C device that this is a 5 V source; with a USB-A socket they are not connected (fit them anyway).
USB-C: VBUS is always on (no "cold socket"), so never connect J11 to another 5 V source (charger, PC).
U12 (ESD array, SOT-23-6) is optional.

J11: one footprint for a USB-A or a USB-C receptacle; fit one of them
USB host port (mouse, joystick, storage) on the Pico USB test points TP2/TP3

Sheet: /USB host port/
File: usb.kicad_sch

Title: PicoGUS THT – USB host port

Size: A4	Date: 2026-10-18	**Rev: v1.3.1**
KiCad E.D.A. 10.0.6		Id: 4/4